Initial Project Planning Template

Date	15 July 2026
Team ID	XXXXXX
Project Name	FloodGuard - AI/ML Flood Prediction System
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation

Sprint	Functional Req. (Epic)	User Story #	User Story / Task	Story Points	Priority
Sprint-1	Data Pipeline	USN-1	As a developer, I can load and clean flood_dataset.xlsx so the data is ready for training	2	High
Sprint-1	Data Pipeline	USN-2	As a developer, I can generate EDA charts (class balance, correlation heatmap) automatically	1	Medium
Sprint-1	Model Training	USN-3	As a developer, I can train Decision Tree, Random Forest, KNN and XGBoost and compare metrics	3	High
Sprint-1	Model Training	USN-4	As a developer, I can automatically select and persist the best model by F1-score	2	High
Sprint-2	Backend API	USN-5	As a developer, I can build a Flask app factory with SQLAlchemy models matching the ERD	3	High
Sprint-2	Backend API	USN-6	As a developer, I can expose a POST /api/predict endpoint that returns flood probability	2	High
Sprint-2	Backend API	USN-7	As a developer, I can log every prediction to the database	2	Medium
Sprint-3	Frontend UI	USN-8	As a user, I can enter weather data through a responsive form	2	High
Sprint-3	Frontend UI	USN-9	As a user, I can see my prediction result dynamically without a page reload	3	High
Sprint-3	Frontend UI	USN-10	As a user, I can view a history of past predictions	2	Medium
Sprint-4	Deployment	USN-11	As a developer, I can push the project to GitHub and deploy it on Render	2	High
Sprint-4	Deployment	USN-12	As a developer, I can fix Python-version/build issues so the app deploys reliably	2	High

Sprint Schedule Summary

Sprint	Sprint Start Date	Sprint End Date (Planned)	Story Points Committed	Team Members
Sprint-1	01 July 2026	05 July 2026	8	Sai Ganesh
Sprint-2	06 July 2026	09 July 2026	7	Sai Ganesh
Sprint-3	10 July 2026	11 July 2026	7	Sai Ganesh
Sprint-4	12 July 2026	12 July 2026	4	Sai Ganesh